

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 15

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Topic

Final Project

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Final Project. State all assumptions and conventions.

$$Registry_{final} = O_1, \dots, O_{52}, O_*$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$L_{53} < L_{accept}$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$hash(\theta_{final}, log, registry) = submission_id$$

Show enough intermediate work that another student can reproduce the calculation.